Dataset Description

VinBigData Chest X-ray Abnormalities Detection Dataset

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1. Introduction

The **VinBigData Chest X-ray Abnormalities Detection Dataset** is a large-scale medical imaging dataset developed by VinBigData and released through a public competition hosted on Kaggle.

The dataset was created to advance research in **automated chest radiograph interpretation**, specifically focusing on **object detection of thoracic abnormalities** using deep learning techniques.

Unlike traditional classification datasets, this dataset is designed for **localization-based detection**, where models must both:

- Identify the presence of abnormalities
- Precisely locate them using bounding boxes

This makes it highly valuable for real-world clinical AI systems aimed at supporting radiologists in high-volume diagnostic environments.

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2. Dataset Objective

The primary objective of this dataset is to develop AI models capable of:

- Detecting radiological abnormalities in chest X-ray images
- Localizing findings with bounding boxes
- Differentiating between multiple pathological conditions
- Identifying normal scans ("No finding")

Such systems can significantly reduce diagnostic workload and assist in early disease detection.

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3. Dataset Composition and Statistics

Attribute	Details
Data Source	VinBigData Institute – Kaggle Competition

Attribute	Details
Image Format	DICOM (.dicom)
Converted Format (Project Use)	PNG
Total Training Images	15,000
Total Test Images	3,000
Image Resolution (Original)	Up to 3000 × 4000 pixels
Preprocessed Resolution	512 × 512 pixels
Annotation Type	Bounding Boxes
Annotation Format (Training)	CSV
Model Training Format	YOLO
Total Classes	14 Abnormalities + 1 "No finding"
Radiologists per Image	3 Independent Experts
Images Used in This Project	50 (Development Phase)

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4. Dataset Structure

The Kaggle dataset is organized into the following structure:

train/

test/

train.csv

sample_submission.csv

Folder Details

- **train/** – Contains 15,000 training DICOM chest X-ray images
- **test/** – Contains 3,000 test DICOM images
- **train.csv** – Annotation file with bounding box coordinates
- **sample_submission.csv** – Format guideline for Kaggle submissions

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5. Annotation File (train.csv) – Detailed Explanation

The train.csv file contains detailed annotations provided independently by three radiologists per image. Each row corresponds to a single finding annotated by one radiologist.

Column Descriptions

1. image_id

- Type: Categorical
- Unique identifier for each chest X-ray
- Matches the DICOM filename (without extension)
- Primary key for linking images and annotations

2. class_name

- Type: Categorical
- Name of the detected abnormality
- Includes 14 disease classes + "No finding"

3. class_id

- Type: Integer
- Encoded numerical label for each class
- Ranges from 0 to 14
- Used during YOLO model training

4. rad_id

- Type: Categorical
- Radiologist identifier (e.g., R1, R2, R3)
- Enables analysis of inter-annotator agreement

5. x_min, y_min

- Type: Float
- Coordinates of top-left corner of bounding box
- Based on original image resolution

6. x_max, y_max

- Type: Float
- Coordinates of bottom-right corner

- Define spatial region of abnormality

During preprocessing, these coordinates are:

1. Scaled to match resized image dimensions (512×512)
2. Converted into normalized YOLO format

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6. Pathological Classes

The dataset contains 14 radiological abnormalities commonly observed in chest radiographs.

Class ID	Class Name	Description
0	Aortic enlargement	Enlargement of the aortic shadow
1	Atelectasis	Lung collapse (partial/complete)
2	Calcification	Calcium deposits in tissues
3	Cardiomegaly	Enlarged cardiac silhouette
4	Consolidation	Alveolar space filled with fluid
5	ILD	Interstitial Lung Disease
6	Infiltration	Hazy opacities from infection/inflammation
7	Lung Opacity	General increase in lung density
8	Nodule/Mass	Discrete round lung lesion
9	Other lesion	Miscellaneous abnormalities
10	Pleural effusion	Fluid in pleural cavity
11	Pleural thickening	Thickened pleural lining
12	Pneumothorax	Air in pleural space
13	Pulmonary fibrosis	Lung tissue scarring
14	No finding	Normal scan (no bounding box)

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7. YOLO Annotation Format

For object detection training, annotations were converted into YOLO format.

Each image has a corresponding .txt file.

YOLO Format:

class_id center_x center_y width height

- All values (except class_id) are normalized between 0 and 1
- Coordinates are relative to image dimensions
- Images labeled as **"No finding"** have empty label files

Conversion Steps:

1. Resize image to 512 × 512
2. Scale bounding box coordinates
3. Convert corner coordinates to center-based format
4. Normalize by dividing with image width/height

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8. Data Preprocessing Workflow

To make the dataset suitable for deep learning training:

- DICOM images converted to PNG
- Pixel intensity normalization applied
- Images resized to fixed resolution
- Bounding boxes scaled accordingly
- Data split for development/testing
- Only 50 images used for initial experimentation

This ensured:

- Reduced computational cost
- Standardized training input
- Faster experimentation cycle

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9. Clinical and Research Significance

This dataset is particularly valuable because:

- Multi-label object detection setting
- Real radiologist annotations

- High-resolution medical images
- Multiple expert reviews per image
- Suitable for transfer learning experiments

Potential applications include:

- Automated triage systems
- Screening assistance in rural healthcare
- Radiologist decision-support systems
- Medical AI research benchmarking

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10. Ethical Considerations

- No personally identifiable patient data included
- Images anonymized before release
- Dataset intended strictly for research and educational use
- Small subset used for demonstration in this project

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11. Project Relevance

For this internship project, the dataset provided:

- A real-world medical AI use case
- Experience with medical image preprocessing
- Object detection pipeline implementation
- Practical exposure to YOLO-based annotation systems

By converting raw medical images into a structured training-ready format, the dataset enabled systematic development of an abnormality detection model.

Final Summary

The VinBigData Chest X-ray dataset offers a comprehensive foundation for developing AI-driven medical image analysis systems. Its multi-expert annotations, bounding box localization framework, and clinically meaningful categories make it a robust benchmark dataset for thoracic abnormality detection.

This project leverages a curated subset of the dataset to demonstrate preprocessing, annotation conversion, and detection workflow development in a structured and research-oriented manner.